

CHAPTER 18

GROUND SUPPORT EQUIPMENT

CHAPTER OVERVIEW

SECTION	TITLE
18-1	Equipment Description and Data
18-2	Troubleshooting
18-3	On Aircraft Inspections
18-4	Maintenance Procedures
18-5	Maintenance Procedures (BENCH)

18-1/(18-2 Blank)

SECTION 18-1

EQUIPMENT DESCRIPTION AND DATA

SECTION OVERVIEW

PARAGRAPH	TITLE
18-1-1	Ground Support Equipment Description and Data

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18-1-1 GROUND SUPPORT EQUIPMENT DESCRIPTION AND DATA.

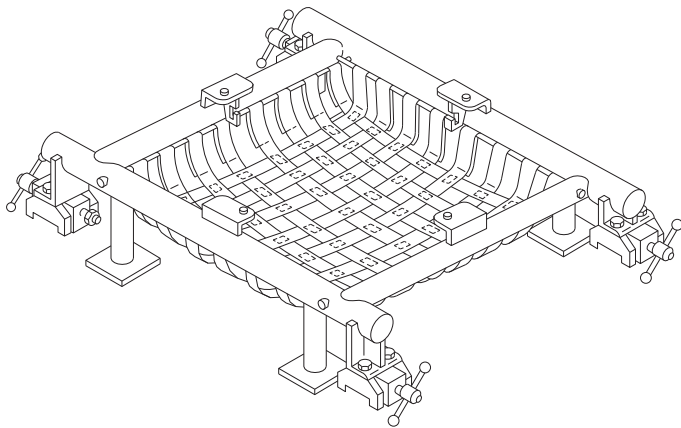
18-1-1.1 Component Handling Adapter Kit.

The component handling adapter kit (Figure 18-1-1-1) consists of six maintenance stand adapters for major components. The major components for these adapters are: main rotor head, main gear box, auxiliary power unit, oil cooler blower, and main and tail rotor blades. Each adapter is easily attached to a standard maintenance trailer by four vise-like latches.

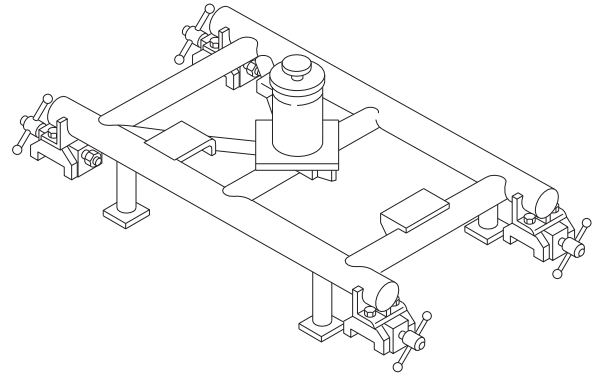
test set to adjust fault isolate fuel quantity system. The harness adapter contains a FUNCTION tank select switch to enable fuel quantity test set to read tank unit capacitance or simulate tank unit capacitance for fuel quantity system adjustment. Two test jacks provide a means for measuring DC voltage output of signal conditioner.

18-1-1.2 Fuel Quantity System Harness Adapter.

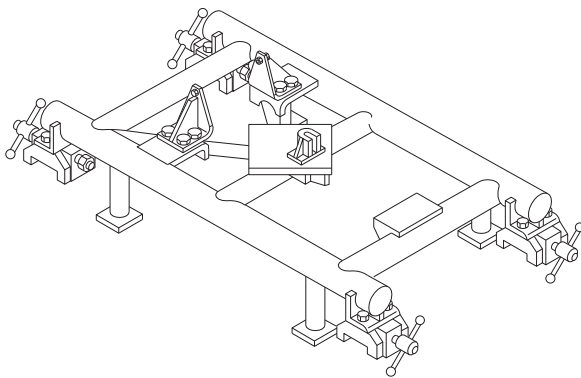
The fuel quantity system harness adapter (Figure 18-1-1-2) is used in conjunction with fuel quantity



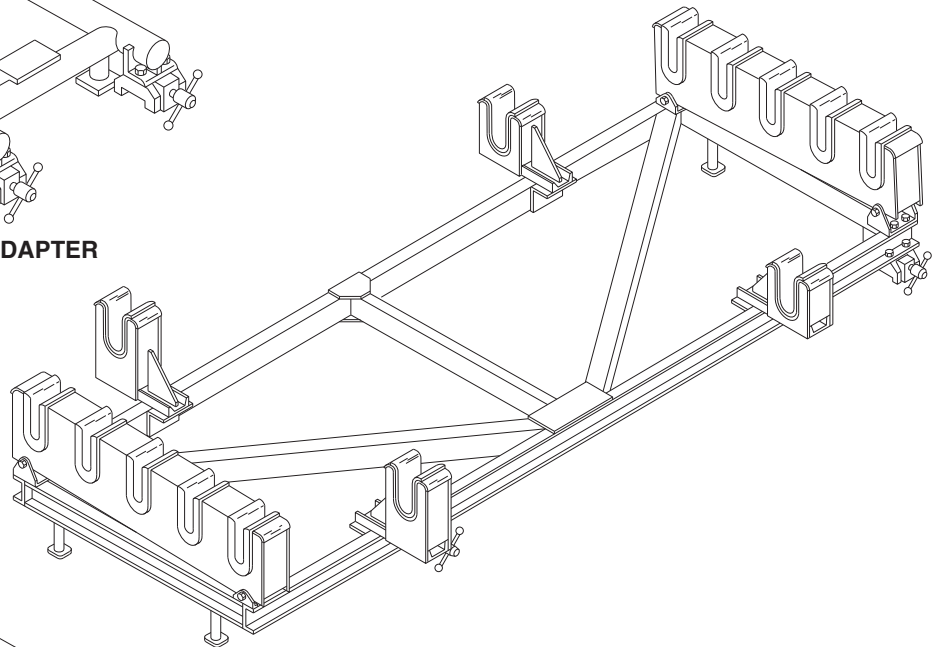
MAIN GEAR BOX ADAPTER
70700-20420-041



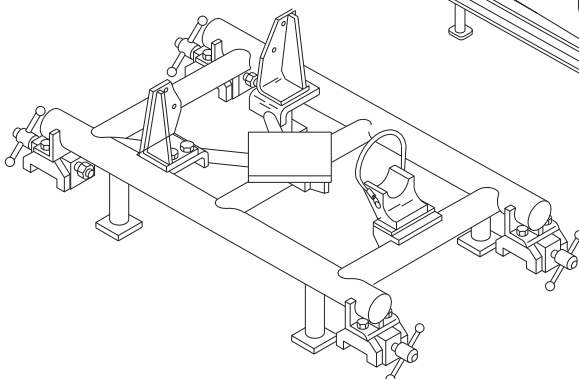
ROTARY WING HEAD ADAPTER
70700-20421-041



AUXILIARY POWER UNIT ADAPTER
70700-20423-041



MAIN AND TAIL ROTOR BLADE ADAPTER
70700-20426-041



TRANSMISSION OIL COOLER ADAPTER
70700-20425-041

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FIGURE 18-1-1-1. COMPONENT HANDLING ADAPTER KIT

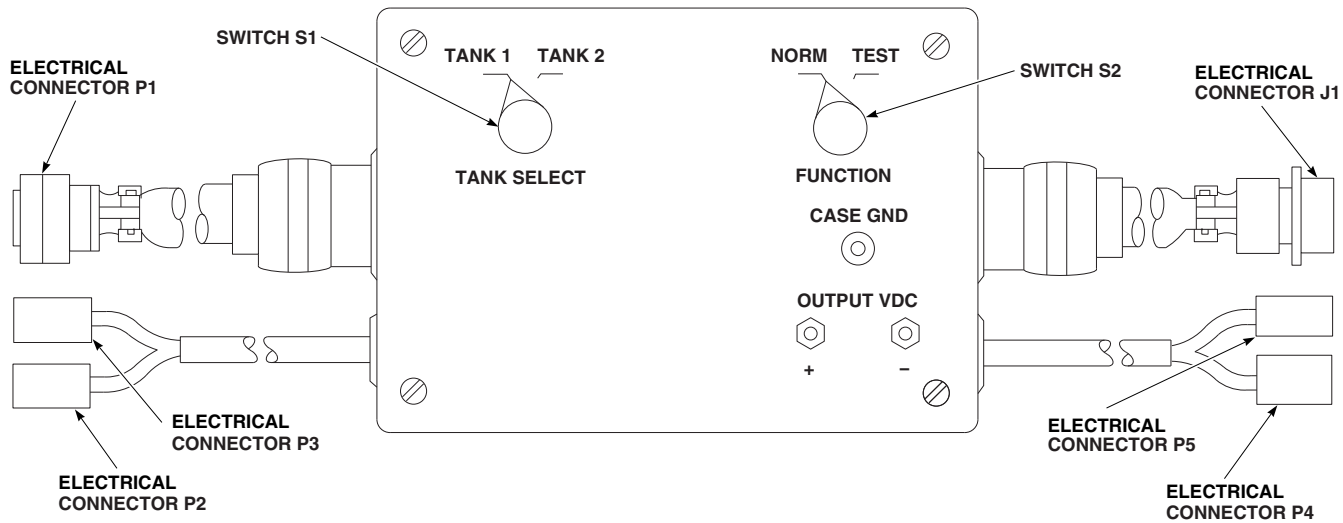


FIGURE 18-1-1-2. FUEL QUANTITY SYSTEM HARNESS ADAPTER

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SECTION 18-2
TROUBLESHOOTING
NOT APPLICABLE

SECTION 18-3
ON AIRCRAFT INSPECTIONS
NOT APPLICABLE

SECTION 18-4

MAINTENANCE PROCEDURES

SECTION OVERVIEW

PARAGRAPH	TITLE
18-4-1	Cabin Maintenance Crane
18-4-2	Tail Maintenance Crane
18-4-3	Maintenance Light
18-4-4	Buddy Start System Hose

18-4-1 CABIN MAINTENANCE CRANE.

PARAGRAPH BREAKDOWN

		PAGE			PAGE
18-4-1.1	Erect Cabin Maintenance Crane	18-4-1-1	18-4-1.2	Inspect/Repair Cabin Maintenance Crane	18-4-1-7

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Cabin Maintenance Crane
- Cargo Strap

Materials/Parts

- Acetone, Item 25, Appendix D
- Grease, Item 147, Appendix D

- Scouring Pad, Item 270, Appendix D

References

- Appendix D
- PARA 2-4-45
- PARA 4-4-4
- PARA 16-4-21
- PARA 16-4-54

18-4-1.1 Erect Cabin Maintenance Crane.



Use of cabin maintenance crane is prohibited when eyebolts, 70209-02136-102 or 70209-02136-104 (upper attachment to airframe), are installed.

18-4-1.1.1 Prepare Helicopter.

- ESSS** If installed on helicopter, perform the following:
 - Remove ESSS external 230 gallon fuel tanks (inboard and outboard) from left or right side of helicopter (PARA 16-4-54).

- Remove upper and lower ESSS horizontal stores support (HSS) root fairings (PARA 16-4-21).
- Remove front and rear vertical support pylon fairings from inboard and outboard HSS (PARA 16-4-21).
- Disconnect the following from inboard ejector rack, BRU-22A/A, adapter:
 - Pneumatic hose.
 - Fuel hose.
 - Electrical connector, J835.
 - Electrical connector, J707.
 - Electrical connector, P709/J709, and remove harness from clamps.

- (f) Bonding wires, two each.
 - (g) Remove rear inboard and outboard quick-release pins and hinge down ejector rack adapter.
- (5) Disconnect the following from outboard ejector rack, BRU-22A/A, adapter:
- (a) Pneumatic hose.
 - (b) Fuel hose.
 - (c) Electrical connector, J849.
 - (d) Electrical connector, J705.
 - (e) Bonding wires, two each.
 - (f) Remove rear inboard and outboard quick-release pins and hinge down ejector rack adapter.
- b. Remove passenger's seats between STA 288.0 and STA 343.0 (PARA 2-4-45).
 - c. Remove screw and washer securing disc plug to soundproofing panel in cabin ceiling at STA 312.3 (Figure 18-4-1-1, Sheet 3, Detail E).
 - d. If engine is to be replaced, remove engine air inlet (PARA 4-4-4).

18-4-1.1.2 Assemble.

WARNING

- Use extreme care when releasing last bolt or nut securing part to be removed on helicopter, as slight out-of-line condition or uneven ground could cause sudden shifting of part.
- Hoist cable may break, causing injury to personnel or damage to equipment if hoist is used to free a sticking component being removed from heli-

copter. Do not use too much force on hoist when removing a component.

- Hoist cable may be damaged or broken, causing injury to personnel or damage to equipment if hook is allowed to contact pulley when removing a component from helicopter. Do not allow hook assembly to contact pulley.

CAUTION

Use of cabin maintenance crane is prohibited when eyebolts, 70209-02136-102 or 70209-02136-104 (upper attachment to airframe), are installed.

NOTE

Cabin maintenance crane may be used on either side of helicopter if rescue hoist is not installed. If rescue hoist is installed, cabin maintenance crane can only be installed on left side of helicopter.

- a. Secure floor beam to passenger's seat studs on cabin floor with quick-disconnects and quick-release pins (Figure 18-4-1-1, Sheet 1, Detail A). Make sure beam is securely attached to floor with quick-disconnects and quick-release pins.
- b. Find proper position where block assembly is to be installed on boom (Figure 18-4-1-1, Sheet 1, Detail B and Table 18-4-1-1). Secure block assembly to block assembly fitting with quick-release pin (Figure 18-4-1-1, Sheet 1, Detail B).
- c. If boom is to be used to replace main transmission, input module, main rotor head, or near engine, stow boom extension as shown (Figure 18-4-1-1, Sheet 2, Detail C).
- d. Connect lower gear box to floor beam with quick-release pin (Figure 18-4-1-1, Sheet 1, Detail B). Allow lower gear box to rest on ground.

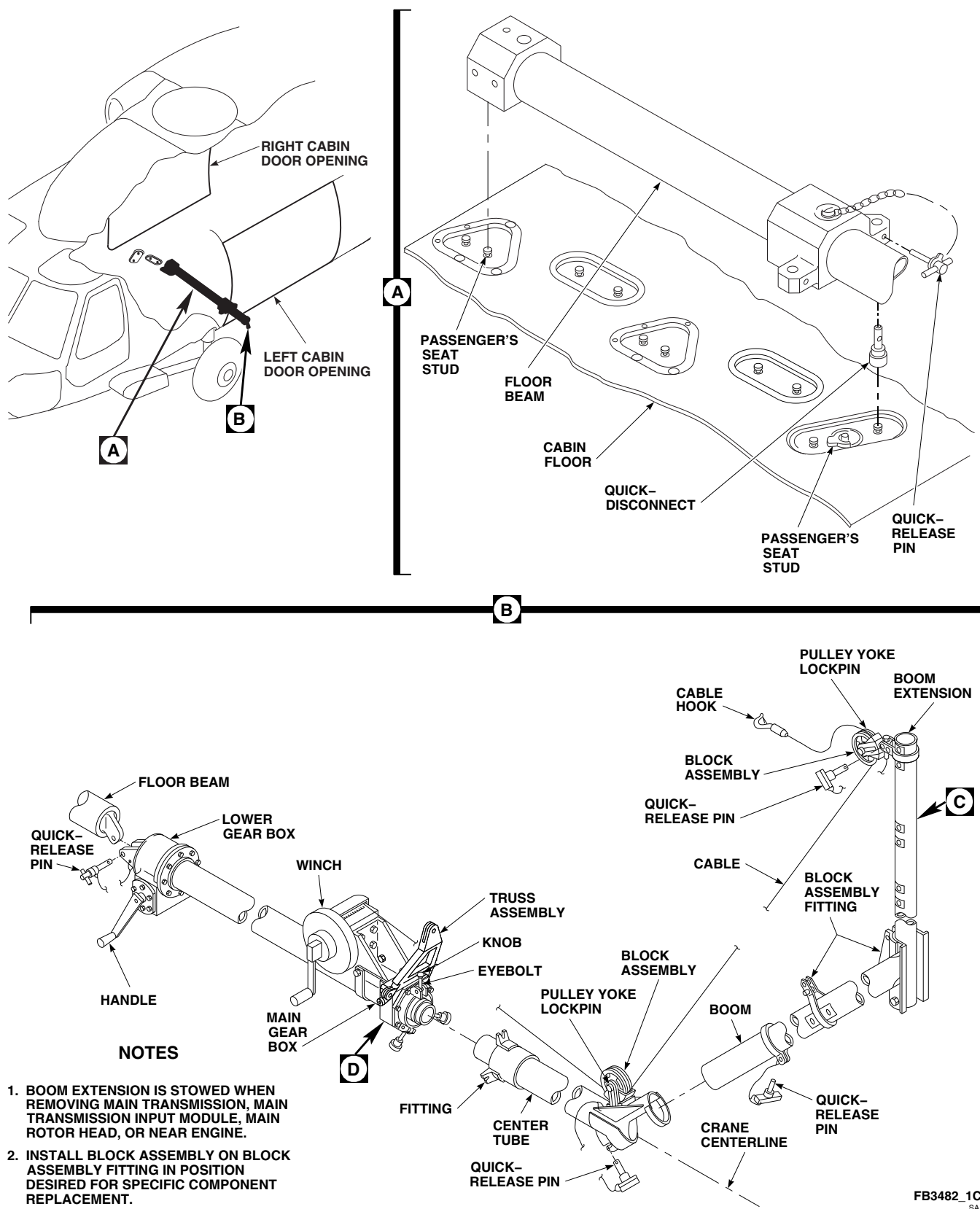


FIGURE 18-4-1-1. ERECT CABIN MAINTENANCE CRANE (SHEET 1 OF 3)

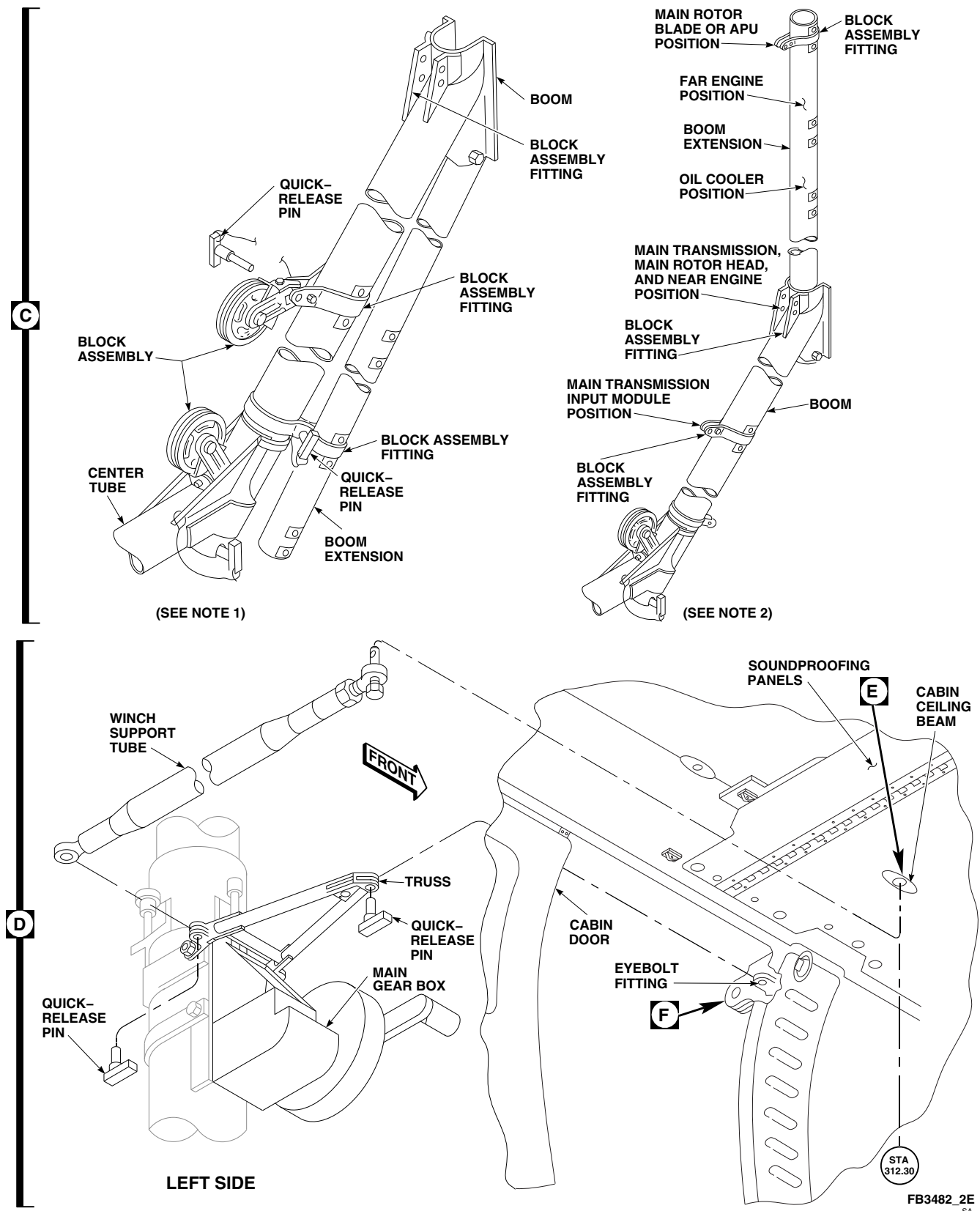


FIGURE 18-4-1-1. ERECT CABIN MAINTENANCE CRANE (SHEET 2 OF 3)

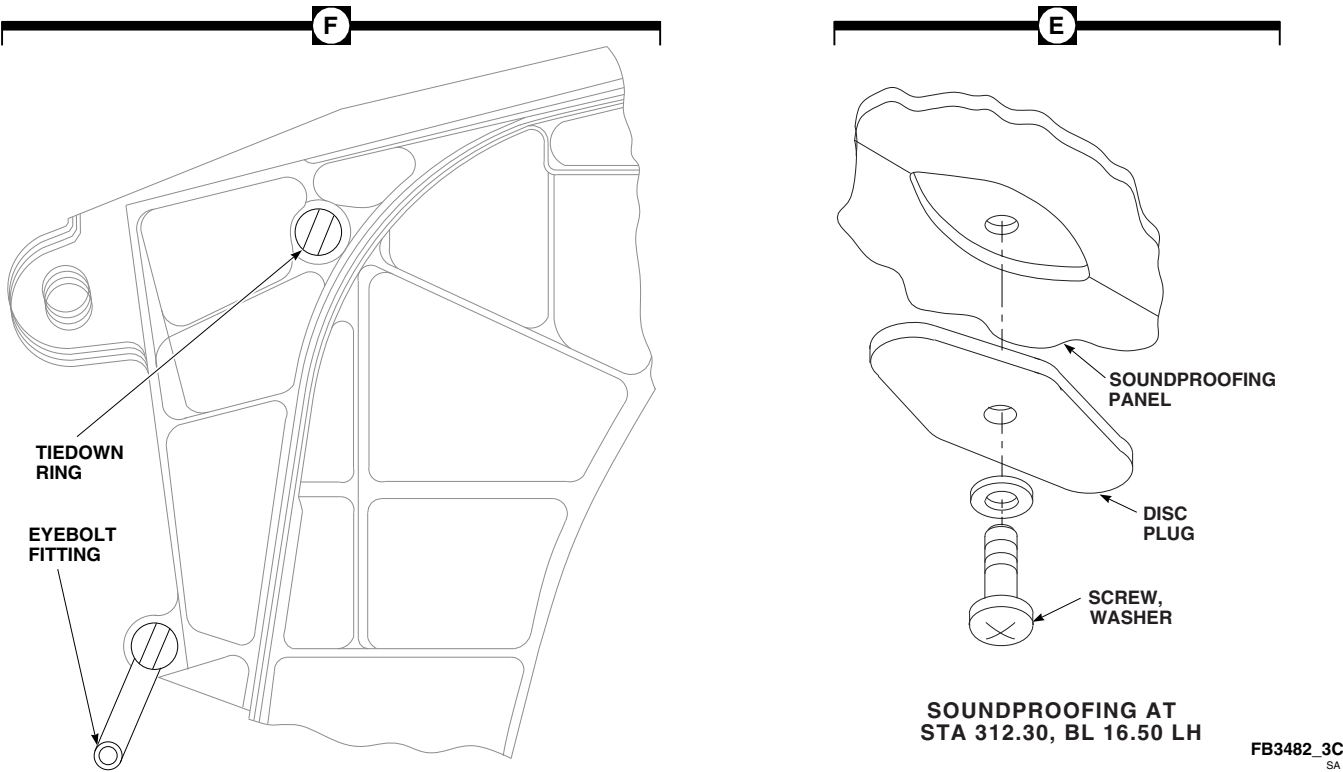


FIGURE 18-4-1-1. ERECT CABIN MAINTENANCE CRANE (SHEET 3 OF 3)

Table 18-4-1-1. Cabin Maintenance Crane Hoisting Data

Component	Approximate Location of Block Assembly (See Note 1)		Approximate Distance from Crane Centerline to Block Assembly (See Note 2)
	STA	BL	Inches
Main Rotor Blade	362.0	154.0 left	110.0
Main Rotor Head	342.0	0.0	52.0
Bifilar	342.0	0.0	52.0
Swashplate	342.0	0.0	52.0
Main Transmission	342.0	0.0	52.0
Input Module	317.0	24.0 left or right	28.0
Oil Cooler	391.0	0.0	82.0
APU	428.0	10.0 left	110.0
Engine (left)	375.0	30.0 left	52.0

Table 18-4-1-1. Cabin Maintenance Crane Hoisting Data (Cont)

Component	Approximate Location of Block Assembly (See Note 1)	Approximate Distance from Crane Centerline to Block Assembly (See Note 2)	
	STA	BL	Inches
Engine (right)	375.0	30.0 right	94.0

NOTES

1. Location of block assembly is given in relation to helicopter STA and BL.
2. Distance is measured at a right angle to cabin maintenance crane centerline.

- e. Slide center tube into main gear box. Engage eyebolts in fittings on center tube and tighten knobs.
- f. Slide boom into center tube, turn 90°, and install quick-release pin locking boom to tube.
- g. Remove pulley yoke lockpins on both block assemblies and install cable from winch on pulleys. Install pulley yoke lockpins.

NOTE

Before winching up cabin maintenance crane, make sure floor beam is securely attached to cabin floor.

- h. Secure cable hook to cargo strap attached to bifilar or main rotor head and winch-up cabin maintenance crane vertically.
- i. Install winch support tube on rear of truss with quick-release pin (Figure 18-4-1-1, Sheet 2, Detail D).
- j. Bolt winch support tube to cabin ceiling beam at STA 312.3 about 3 1/2-inches to rear of cargo door opening.
- k. **ESSS** Secure truss assembly to eyebolt fitting.

1. **W/O ESSS** Secure truss assembly to eyebolt fitting on tiedown ring (Figure 18-4-1-1, Sheet 2, Detail D and Sheet 3, Detail F).
- m. Secure truss to eyebolt fitting with quick-release pin.
- n. Turn handle on lower gear box to position boom over component to be removed (Figure 18-4-1-1, Sheet 1, Detail B). Locate block assembly in relation to helicopter STA and BL (Table 18-4-1-1).

18-4-1.1.3 Disassemble.

- a. Remove quick-release pin securing truss to eyebolt fitting (Figure 18-4-1-1, Sheet 2, Detail D and Sheet 3, Detail F).
- b. Remove truss from eyebolt fitting.
- c. Unbolt winch support tube from cabin ceiling beam at STA 312.3 (Figure 18-4-1-1, Sheet 2, Detail D).
- d. Remove quick-release pin and remove winch support tube from rear of truss.
- e. Winch-down cabin maintenance crane horizontally. Disconnect cable hook from cargo strap attached to bifilar or main rotor head.

- f. Remove pulley yoke lockpins and remove cable from winch on pulleys. Replace pulley yoke lockpins.
- g. Remove quick-release pin locking boom to center tube. Slide boom out of center tube and turn 90° (Figure 18-4-1-1, Sheet 1, Detail B).
- h. Loosen knobs on center tube and disengage eyebolts in fittings. Slide center tube out of main gear box.
- i. Remove quick-release pin and disconnect lower gear box from floor beam.
- j. Remove quick-release pin securing block assembly to block assembly fitting.
- k. Remove quick-release pins and disconnect quick-disconnects, and remove floor beam from cabin floor (Figure 18-4-1-1, Sheet 1, Detail A).
- l. If engine was replaced, install engine air inlet (PARA 4-4-4).
- m. Install disc plug in soundproofing panel at cabin ceiling STA 312.3, and secure with screw and washer (Figure 18-4-1-1, Sheet 3, Detail E).
- n. Install passenger's seats between STA 288.0 and STA 343.0 (PARA 2-4-45).
- o. **ESSS** If installed on helicopter, perform the following:
 - (1) Connect the following to outboard ejector rack, BRU-22A/A, adapter:
 - (a) Install rear inboard and outboard quick-release pins and hinge up ejector rack adapter.
 - (b) Bonding wires, two each.
 - (c) Electrical connector, J705.
 - (d) Electrical connector, J849.
 - (e) Fuel hose.

- (f) Pneumatic hose.
- (2) Connect the following to inboard ejector rack, BRU-22A/A, adapter:
 - (a) Install rear inboard and outboard quick-release pins and hinge down ejector rack adapter.
 - (b) Bonding wires, two each.
 - (c) Electrical connector, P709/J709, and install harness to clamps.
 - (d) Electrical connector, J707.
 - (e) Electrical connector, J835.
 - (f) Fuel hose.
 - (g) Pneumatic hose.
- (3) Install front and rear ESSS horizontal stores support (HSS) vertical support pylon fairings to inboard and outboard stores support (PARA 16-4-21).
- (4) Install upper and lower ESSS horizontal stores support (HSS) root fairings (PARA 16-4-21).
- (5) Install ESSS external 230 gallon fuel tanks (inboard and outboard) to left or right side of helicopter (PARA 16-4-54).

18-4-1.2 Inspect/Repair Cabin Maintenance Crane.

- a. Check all tubes for dents, cracks, and corrosion. **DENTS MUST NOT EXCEED ¼-INCH IN DEPTH. NO CRACKS OR CORROSION ALLOWED.** If dents exceed limit, replace tube. If cracked, weld cracks. Light corrosion can be cleaned using scouring pad, Item 270, Appendix D. If tubes have heavy corrosion, replace tubes.
- b. Check all quick-release pins and fittings for breaks, cracks, and corrosion. **NONE ALLOWED.** If broken, cracked, or corroded, replace pins and fittings.
- c. Check block assemblies for wear, binding, and cracks. **WEAR ON BLOCK ASSEMBLY**

SHALL NOT EXCEED 1/8-INCH. PLAY IN BEARINGS SHALL NOT EXCEED 0.020-INCH IN ANY DIRECTION. NO CRACKS ALLOWED. If damage exceeds limits, replace block assembly.

- d. Check winch support tube for dents and play in bearings. **NONE ALLOWED.** If dent(s) or play is found, replace bearing(s) as follows (Figure 18-4-1-2, Detail A):
 - (1) Remove pin from non-adjustable end and remove bearing from winch support tube.
 - (2) Remove special bolt from bearing by removing retaining ring and collars.
 - (3) Loosen jamnut on adjustable end. Unscrew bearing from tube. Remove jamnut and quick-release pin from bearing.
 - (4) Place special bolt and collars on bearing and secure with retaining ring. **PLACE QUICK-RELEASE PIN IN BEARING. THREAD BEARING INTO TUBE UNTIL MEASUREMENT BETWEEN BEARING CENTERS IS 31.89-INCHES.** Lock bearing in place using jamnut.
 - (5) Install bearing in tube and secure with pin.
 - (6) Place cable assembly over other bearing and thread jamnut on as far as possible.
- e. Check truss for corrosion, dents, and wear. **NONE ALLOWED.** If truss is corroded, dented, or worn, remove all quick-release pins and replace truss (Figure 18-4-1-2, Detail B).
- f. Check winch for smooth operation and check cable for condition. **WINCH MUST NOT BIND OR SLIP UNDER LOAD. NO MORE THAN TWO STRANDS OF CABLE MAY BE BROKEN. OUTER STRANDS OF CABLE MUST NOT BE BLENDED TOGETHER, DUE TO WEAR.** If any of the above conditions are met or exceeded, replace winch as follows (Figure 18-4-1-2, Detail C):

NOTE

- Winch and cable are replaced as one unit.
- Note screw location before removing.
 - (1) Remove screws securing winch to winch support.
 - (2) Install replacement winch on winch support and secure with screws.
- g. Check winch support for cracks, dents, and corrosion. **NONE ALLOWED.** If winch support is cracked, dented, or corroded, replace by removing bolts and washers securing it to lower tube assembly.
- h. Check bearings on lower tube assembly for play. **NONE ALLOWED.** If bearings have play, replace bearings as follows (Figure 18-4-1-2, Detail D):
 - (1) Remove locknut. Slide off coupling and key.
 - (2) Remove bolts and washers securing cover plate to housing. Remove cover plate.
 - (3) Remove housing from lower tube assembly. Remove upper bearing and spacer from housing.
 - (4) Remove screws, retainer, and lower bearing from housing.
 - (5) Using acetone, Item 25, Appendix D, clean all components.
 - (6) Inspect for wear and damage. **NONE ALLOWED.** If damaged, replace parts as necessary.
 - (7) Using grease, Item 147, Appendix D, repack bearings.
 - (8) Place lower bearing in housing with seal facing down. Position retainer on housing and secure with screws.
 - (9) Slide housing assembly over lower tube assembly. Slide spacer over tube, seating

it on lower bearing. Using grease, Item 147, Appendix D, fill gap between spacer and housing.

- (10) Install upper bearing with seal facing up.
- (11) Install cover plate on housing and secure with bolts and washers.

- (12) Slide coupling and key on tube.

- (13) Install locknut and tighten until no play is felt in bearings.

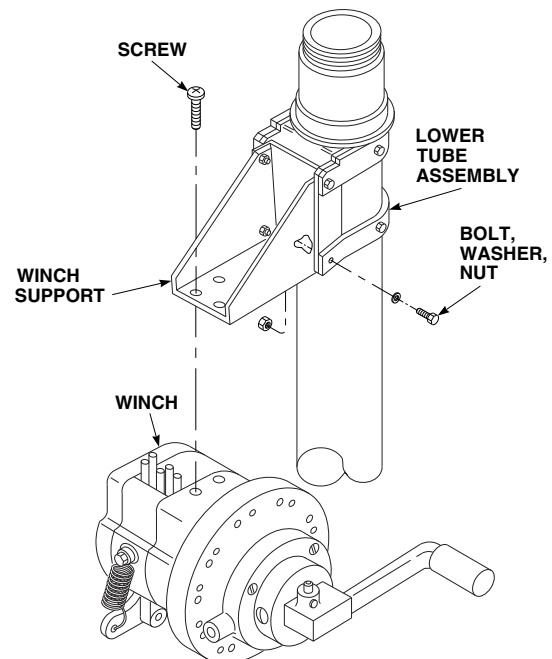
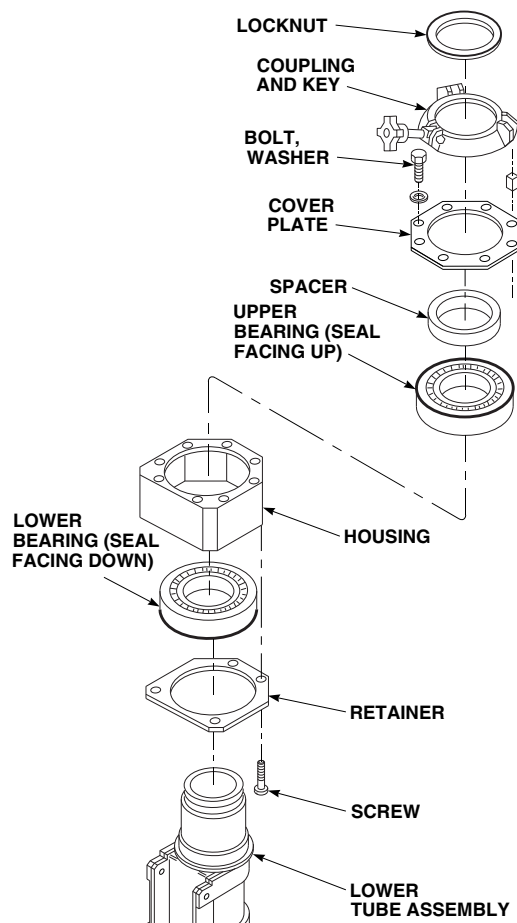
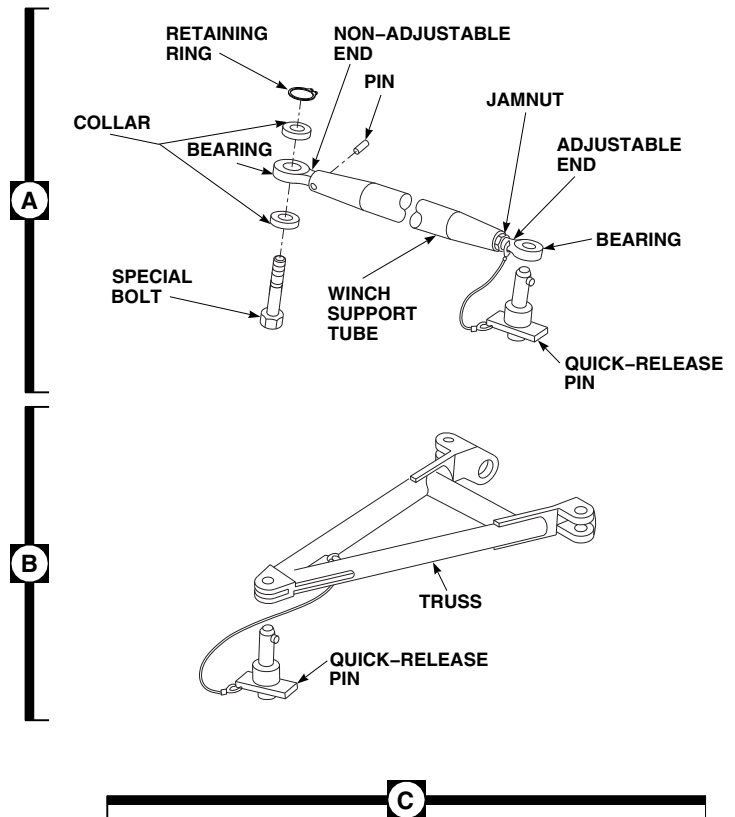
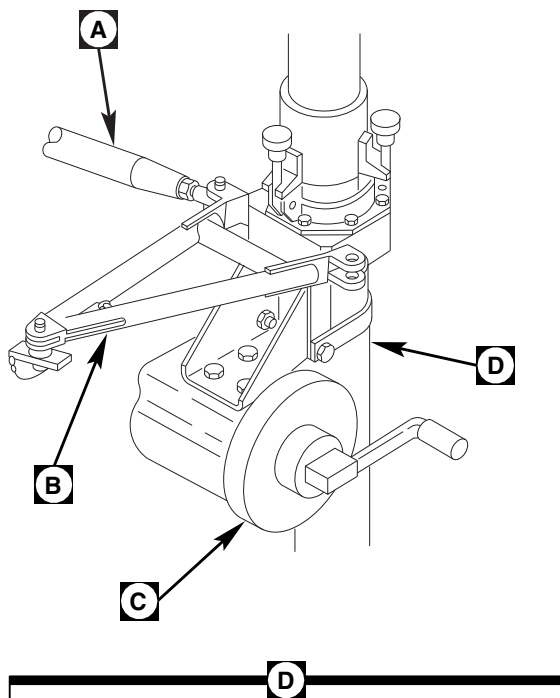


FIGURE 18-4-1-2. INSPECT/REPAIR CABIN MAINTENANCE CRANE

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18-4-2 TAIL MAINTENANCE CRANE.

PARAGRAPH BREAKDOWN

	PAGE
18-4-2.1 Install Tail Maintenance Crane	18-4-2-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Maintenance Platform
- Tail Maintenance Crane

References

- PARA 2-4-137

18-4-2.1 Install Tail Maintenance Crane.

18-4-2.1.1 Prepare Helicopter.

- If tail maintenance crane will be used to remove tail gear box, tail gear box fairings must be removed (PARA 2-4-137).
- If tail maintenance crane will be used to remove tail rotor blades, remove access panel on top of rear tail gear box fairing as follows:

NOTE

Access panel is located behind anticollision light.

- Position suitable maintenance platform next to tail gear box fairing.
- Loosen turnlock fasteners and remove access panel (Figure 18-4-2-1, Detail A).

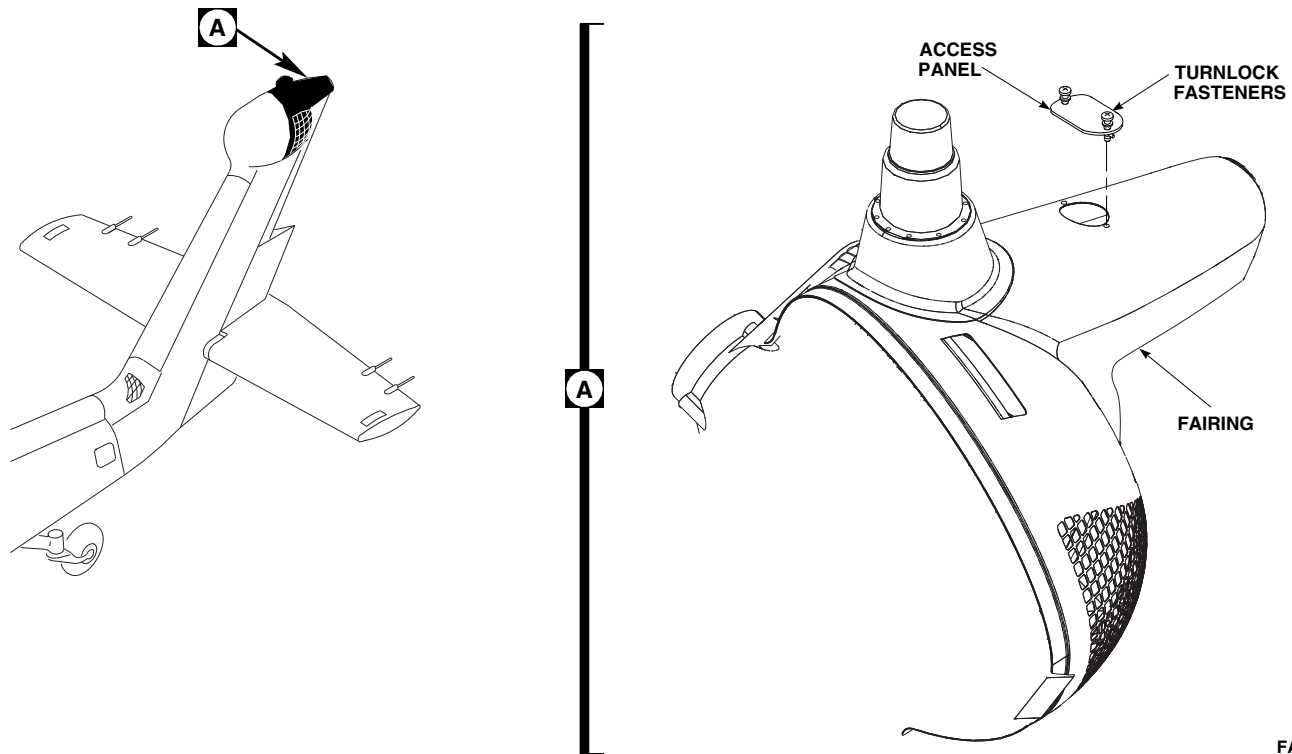
18-4-2.1.2 Install.

- Insert support assembly through hole into tail rotor pylon (Figure 18-4-2-2, Detail).
- Install arm assembly into support assembly, positioning as follows:

- If removing tail gear box, position arm pointing forward.
- If removing tail rotor blades, position arm pointing to the side.
- Lock arm assembly in place with pin.
- Connect chain hoist to arm assembly as follows:
 - If removing tail gear box, install chain hoist on lower lug.
 - If removing tail rotor blades, install chain hoist on upper lug.

18-4-2.1.3 Remove.

- Disconnect chain hoist from arm assembly (Figure 18-4-2-2, Detail A).
- Remove locking pin from arm assembly.
- Remove arm assembly from support assembly.
- Remove support assembly from hole in pylon.

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e. Install tail gear box fairings (PARA 2-4-137) or access panel as follows (Figure 18-4-2-1, Detail A):

(1) Position suitable maintenance platform next to tail gear box.

(2) Position access panel on top of rear tail gear box fairing and secure with turnlock fasteners.

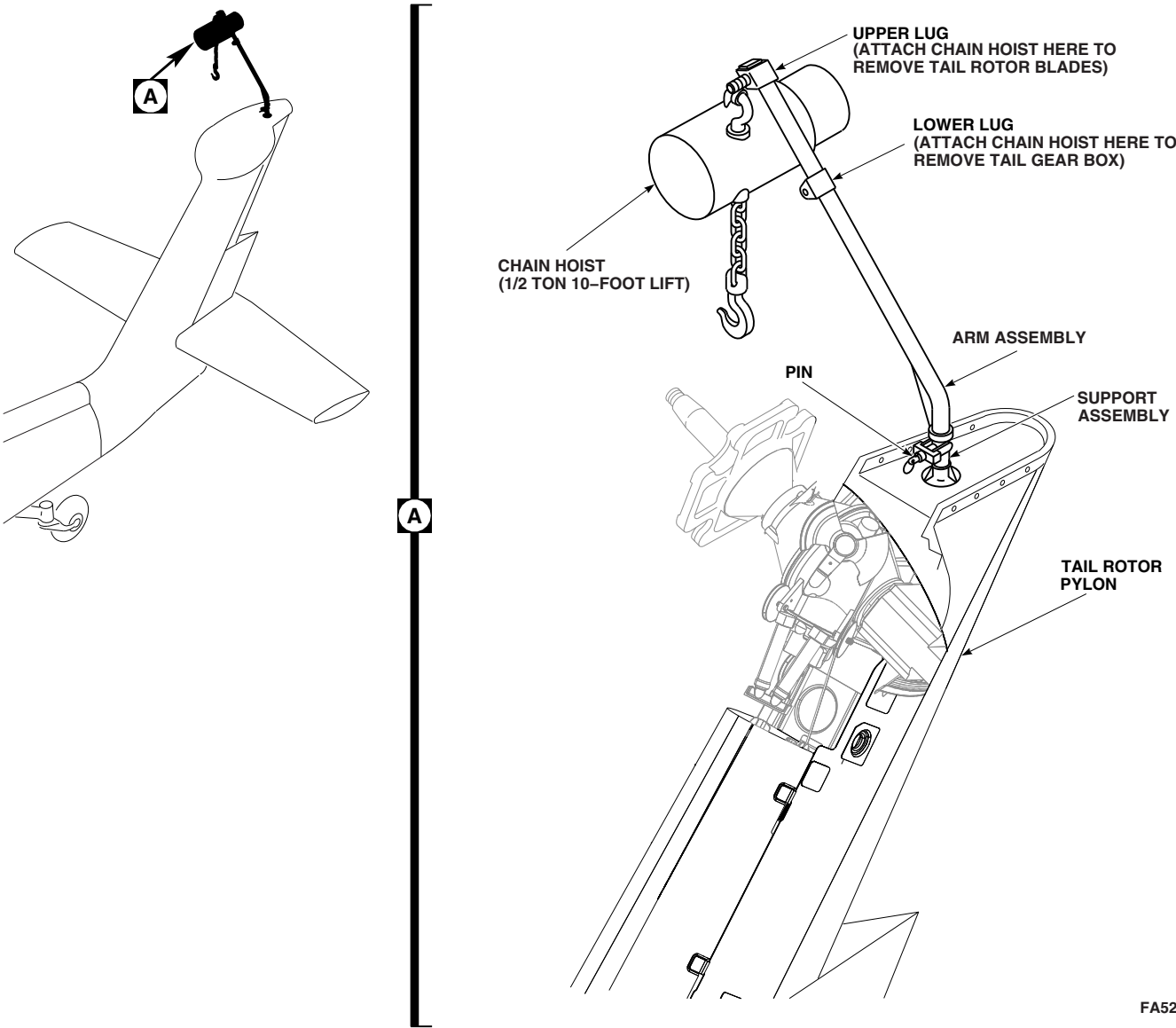


FIGURE 18-4-2-2. INSTALL TAIL MAINTENANCE CRANE

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18-4-3 MAINTENANCE LIGHT.

PARAGRAPH BREAKDOWN

		PAGE
18-4-3.1	Replace Maintenance Light	18-4-3-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit

Materials/Parts

- Protective Caps and Plugs

References

- PARA 1-7-20

18-4-3.1 Replace Maintenance Light.

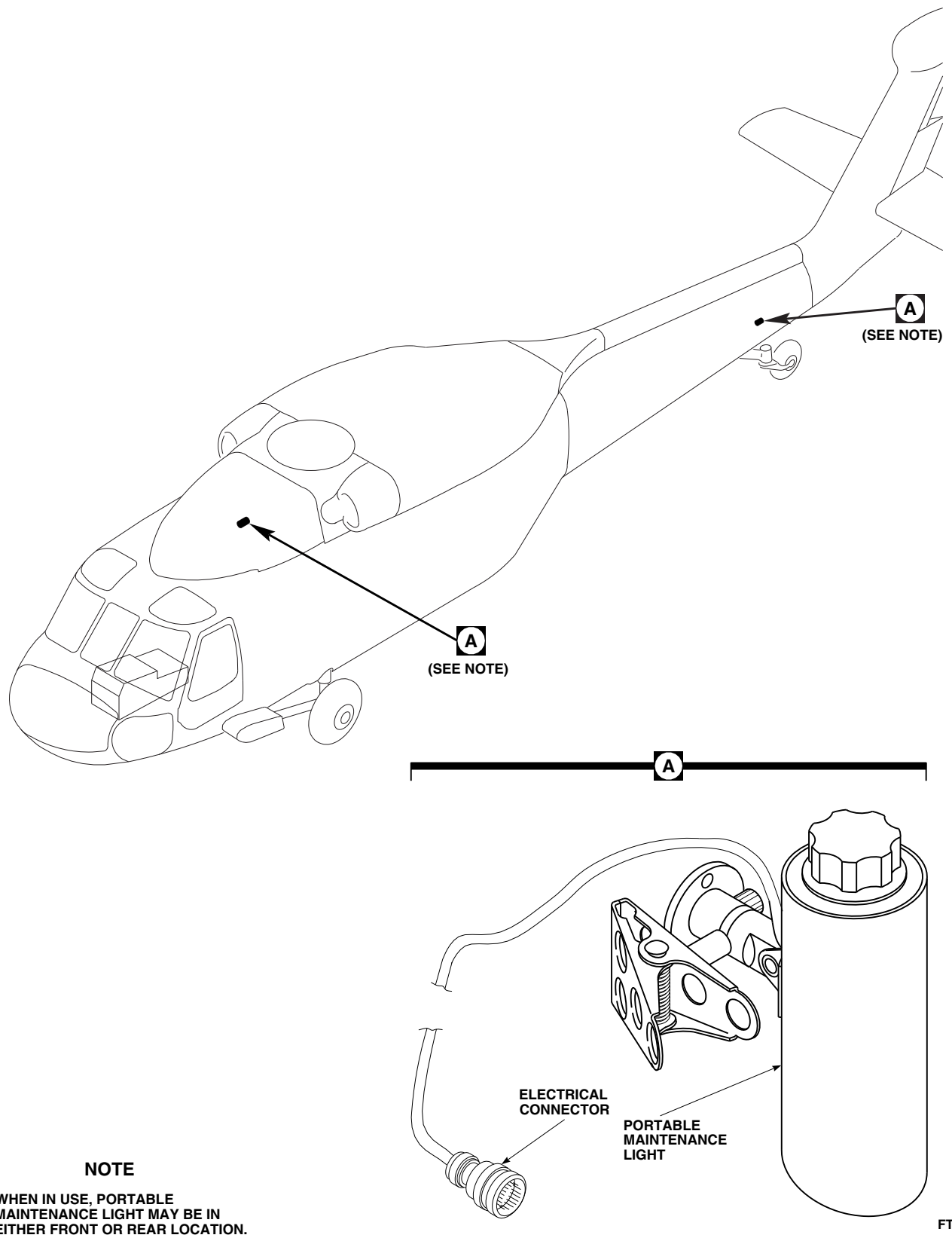
18-4-3.1.1. Install.

- Remove maintenance light from stowage bag on back of pilot’s seat.
- If installing maintenance light in cabin, open access door in soundproofing at STA 293.0 in cabin ceiling on right side of helicopter.
- Remove protective cover from MAINT LIGHT receptacle and protective cap from electrical connector.
- If required, inspect and treat receptacle and electrical connector (PARA 1-7-20).

- Connect light to receptacle (Figure 18-4-3-1, Detail A).

18-4-3.1.2. Remove.

- Disconnect light from MAINT LIGHT receptacle (Figure 18-4-3-1, Detail A).
- Install protective cover on receptacle and protective cap on electrical connector.
- Coil light cord and store light in stowage bag on back of pilot’s seat.
- Close soundproofing access door in cabin ceiling.



18-4-4 BUDDY START SYSTEM HOSE.

PARAGRAPH BREAKDOWN

	PAGE
18-4-4.1 Repair Buddy Start System Hose	18-4-4-1

INITIAL SETUP

- Torque Wrench, 30 - 150 in. lbs

Tools

- Aircraft Mechanic’s Toolkit
- Clamp Tool

18-4-4.1 Repair Buddy Start System Hose.

- (3) Pull coupling assembly from hose.

18-4-4.1.1 Disassemble.

- Remove quick clip and pin by cutting cable at adapter and quick clip. Remove cable securing pin to quick clip (Figure 18-4-4-1, Detail A).
- Remove protective cap by cutting cable at adapter and protective cap.
- Remove adapter as follows:
 - Cut and remove outer clamp.
 - Pull back hose jacket to expose inner clamp. Cut and remove clamp. Remove band.
 - Pull adapter from hose.
- Remove protective shroud by cutting cable at shroud and at handle.
- Remove coupling assembly as follows:
 - Cut and remove outer clamp.
 - Pull back hose jacket to expose inner clamp. Cut and remove inner clamp. Remove band.

18-4-4.1.2 Assemble.

NOTE

Make sure clamp is centered on band.

- Pull hose jacket back and place band around hose at inner clamp location. Place clamp over band (Figure 18-4-4-1, Detail A).
- Insert adapter into hose.
- Tighten and lock clamp as follows:
 - Place tail of clamp into clamp tool and grip tail end with gripper lever.
 - Apply tension to clamp by turning T-bolt clockwise until hose starts to compress. **TORQUE T-BOLT TO 95 - 105 INCH-POUNDS.**
 - Roll tool over hose, backing off tension handle ¼ to ½ turn through course of roll, until shear hook engages T-bolt edge.
 - Tighten knurled nut by hand securing T-bolt tightly between edge of hook and

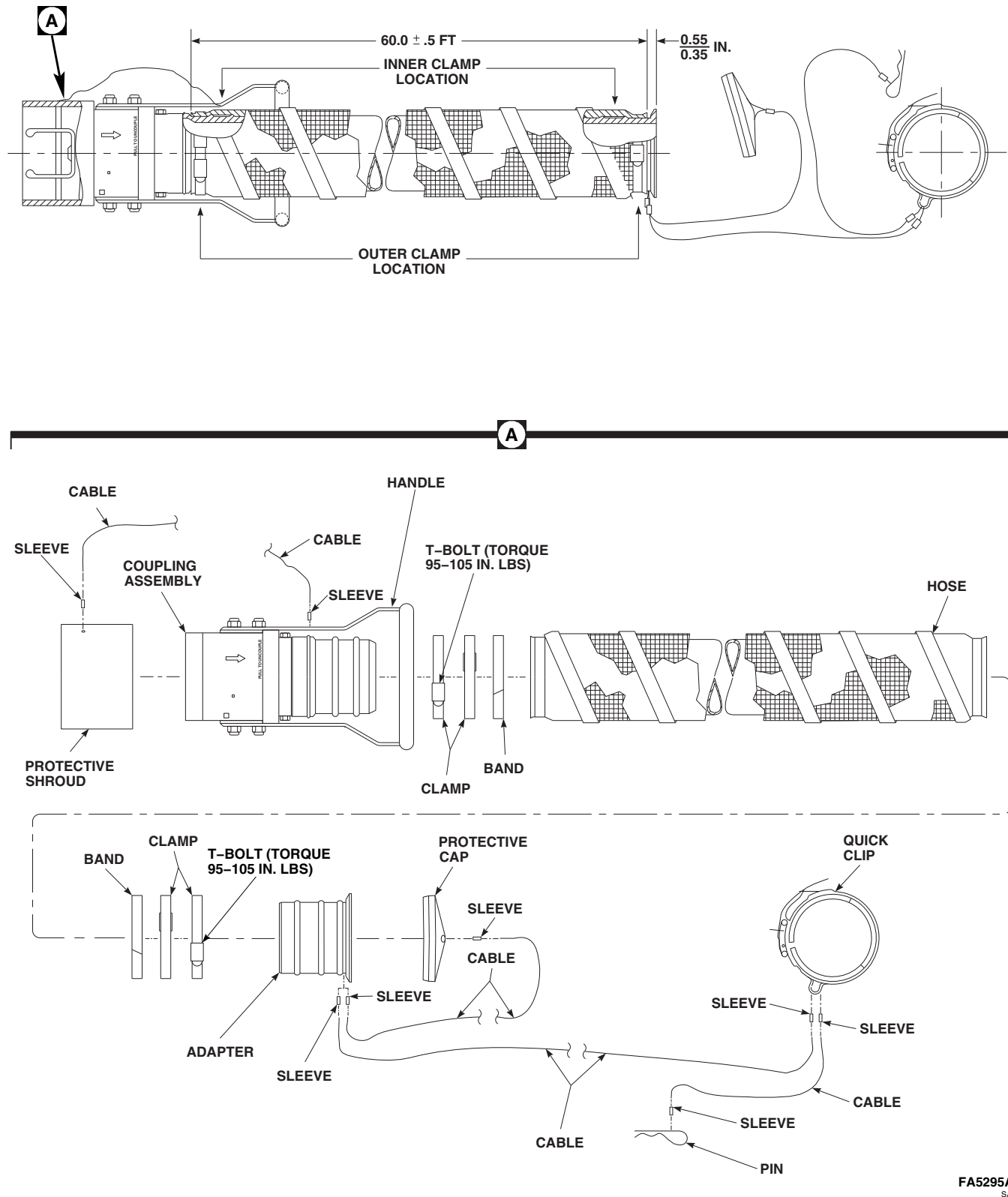


FIGURE 18-4-1. REPAIR BUDDY START SYSTEM HOSE

nose of adapter. Turn hex nut clockwise until band is sheared.

- d. Push hose jacket over clamp and back into place. Place clamp over jacket and hose at outer clamp location. Locate T-bolt 180° opposite from inner clamp T-bolt location.
- e. Tighten and lock clamp in place. Refer to step c.
- f. Install cable to protective cap and adapter.
- g. Install cable to pin and quick clip.
- h. Install cable to quick clip and adapter.
- i. Insert coupling assembly into hose.

NOTE

Make sure clamp is centered on band.

- j. Pull hose jacket back and place band around hose at inner clamp location. Place clamp over band.
- k. Tighten and lock clamp in place. Refer to step c.
- l. Push hose jacket over clamp and back into place. Place clamp over jacket and hose at outer clamp location. Locate T-bolt 180° opposite from inner clamp T-bolt location.
- m. Tighten and lock clamp in place. Refer to step c.
- n. Install cable to protective shroud and handle.

SECTION 18-5

MAINTENANCE PROCEDURES (BENCH)

SECTION OVERVIEW

PARAGRAPH	TITLE
18-5-1	Cabin Maintenance Crane (BENCH)

18-5-1/(18-5-2 Blank)

18-5-1 CABIN MAINTENANCE CRANE (BENCH).

PARAGRAPH BREAKDOWN

	PAGE
18-5-1.1 Repair Cabin Maintenance Crane Gear Box (BENCH)	18-5-1-1

INITIAL SETUP

Tools

- Aircraft Mechanic’s Toolkit
- Feeler Gage
- Phenolic Drift
- Plastic Hammer

Materials/Parts

- Acetone, Item 25, Appendix D
- Adhesive, Item 37, Appendix D
- Grease, Item 147, Appendix D

References

- Appendix D

18-5-1.1 Repair Cabin Maintenance Crane Gear Box (BENCH).

18-5-1.1.1 Disassemble.

- Loosen setscrew and slide handle and key off shaft (Figure 18-5-1-1).
- Remove bolts and washers securing open and closed caps on housing. Remove caps and peel-off shims.
- Check grease seal in open cap. If necessary, replace seal.

NOTE

Wormgear and gear must be replaced as a set.

- Remove wormgear by turning gear to loosen bearing packages. Remove cups and cones from wormgear.

- Remove bolts and washers from bearing retainer, and pull lower tube from housing.
- Remove nut, lower bearing, gear, and key from lower tube.
- Using plastic hammer, tap bearing retainer from lower tube. Remove upper bearing and shim from retainer.
- Remove screws and washers securing cover and gasket to housing.
- Clean all parts in acetone, Item 25, Appendix D.
- Inspect parts for wear and damage. **NONE ALLOWED.** Replace parts as necessary.

18-5-1.1.2 Inspect/Repair.

- Pack all bearings with grease, Item 147, Appendix D, or equivalent.

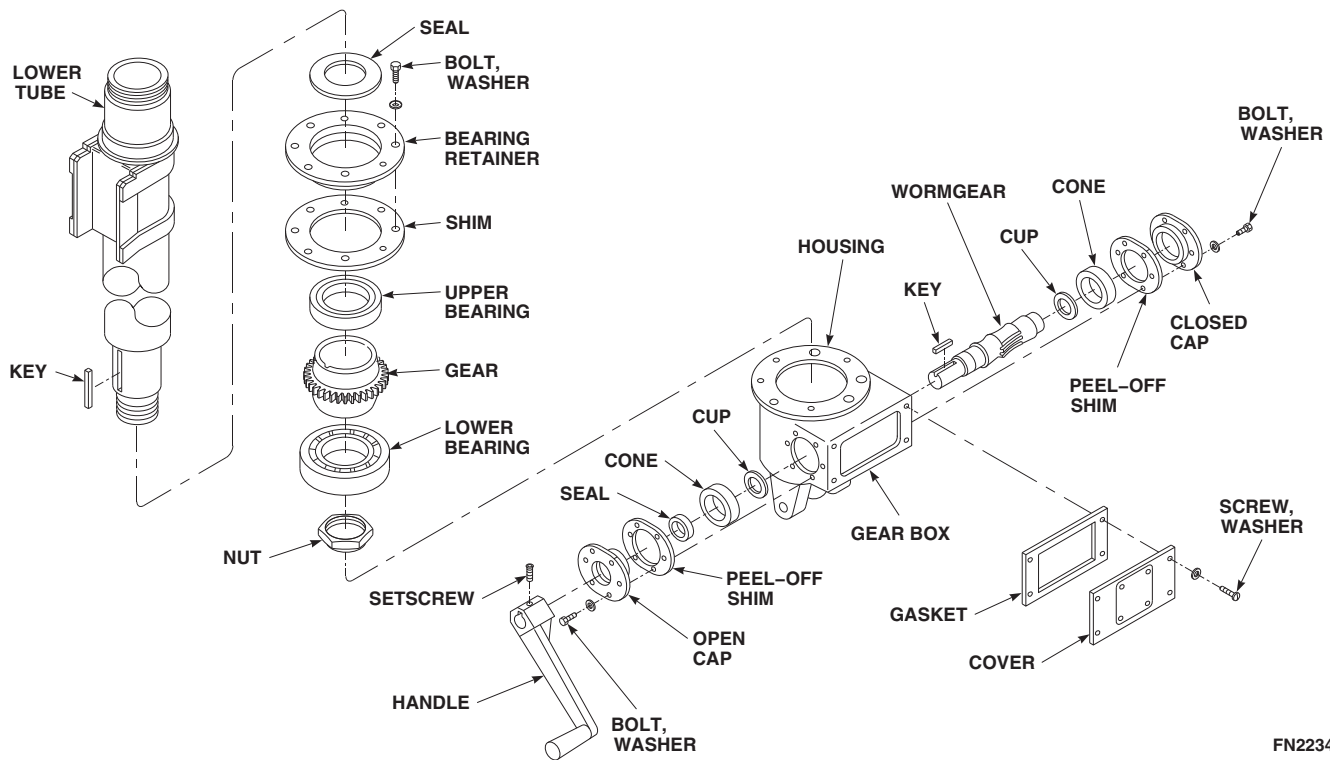
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FIGURE 18-5-1-1. REPAIR CABIN MAINTENANCE CRANE GEAR BOX (BENCH)

- b. Inspect seal on lower tube for damage. **NONE ALLOWED.** If damaged, bond new seal to lower tube using adhesive, Item 37, Appendix D, per manufacturer's instructions.
- c. Press upper bearing into retainer. Bearing seal should be on top after installation.

18-5-1.1.3 Assemble.

- a. Install bearing retainer on lower tube using plastic hammer to seat bearing (Figure 18-5-1-1).
- b. Install gear (unmarked face on top) and key on lower tube, using plastic hammer and phenolic drift to seat it against bearing.
- c. Install lower bearing with seal on bottom. Secure bearing with nut.
- d. Install bearing retainer as follows:
 - (1) Place shim and lower tube on housing.
 - (2) Measure clearance between shim and bearing retainer using feeler gage. **CLEARANCE SHALL BE 0.001-INCH.**

- (3) If necessary, peel shim to obtain proper clearance.
- (4) Secure retainer with bolts and washers.
- e. Press cones on wormgear. Slide wormgear in housing.
- f. Place cups in housing.

NOTE

If wormgear or its bearings were replaced, replace peel-off shims.

- g. Install peel-off shims and caps with bolts and washers. Tighten bolts in pairs opposite to each other to evenly draw covers and cups into housing.
- h. Check that wormgear is centered on gear, and there is no endplay on worm. If endplay is found, perform the following:
 - (1) Remove bolts and washers securing open and closed caps on housing. Remove caps and peel-off shims.

- (2) Peel equal layers of shim to eliminate endplay.
- (3) Install peel-off shims and caps with bolts and washers. Tighten bolts in pairs opposite to each other to evenly draw covers and cups into housing.
 - i. Install handle and secure with key and setscrew.
 - j. Install cover and gasket on housing and secure with screws and washers.

